Lab 03 – Python Operators

Programming Exercise:

Question 1. A ball at the end of a string is revolving uniformly in a horizontal circle of radius 2 meters at constant angular speed 10 rad/s. Determine the magnitude of the linear velocity of a point located:

(a) 0.5 meters from the center
(b) 1 meter from the center
(c) 2 meters from the center

Source Code:

(a)

```
angular_speed = 10

# Part (a): Calculate linear velocity at 0.5 meters from the center
radius_a = 0.5
linear_velocity_a = angular_speed * radius_a
print("Linear velocity at 0.5 meters from the center:", linear_velocity_a,"m/s")
```

(b)

```python
angular_speed = 10

# Part (a): Calculate linear velocity at 0.5 meters from the center
radius_b = 1
linear_velocity_b = angular_speed * radius_b
print("Linear velocity at 1 meters from the center:", linear_velocity_b,"m/s")

```

(c)

```python
angular_speed = 10

# Part (a): Calculate linear velocity at 0.5 meters from the center
radius_c = 2
linear_velocity_c = angular_speed * radius_c
print("Linear velocity at 2 meters from the center:", linear_velocity_c,"m/s")
```

Output:

(a)

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Linear velocity at 0.5 meters from the center: 5.0 m/s

Process finished with exit code 0
```

(b)

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Linear velocity at 1 meters from the center: 10 m/s

Process finished with exit code 0
```

(c)

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Linear velocity at 2 meters from the center: 20 m/s

Process finished with exit code 0
```

Question 2. The blades in a blender rotate at a rate of 5000 rpm. Determine the magnitude of the linear velocity:

(a) a point located 5 cm from the center

(b) a point located 10 cm from the center

Source Code:

(a)

```
angular_speed = 523.3

# Part (a) for Question 2: Calculate linear velocity at 5 cm (0.05 meters) from the center
radius_a = 0.05
linear_velocity_a = angular_speed * radius_a
print(f"Linear velocity at 5 cm (0.05 meters) from the center: {linear_velocity_a} m/s")
```

(b)

```
angular_speed = 523.3

radius_b = 0.1
linear_velocity_b = angular_speed * radius_b
print(f"Linear velocity at 10 cm (0.01 meters) from the center: {linear_velocity_b} m/s")
```

Output:

(a)

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Linear velocity at 5 cm (0.05 meters) from the center: 26.165 m/s

Process finished with exit code 0
```

(b)

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Linear velocity at 10 cm (0.01 meters) from the center: 52.33 m/s


Process finished with exit code 0
```

Question 3. A point on the edge of a wheel 30 cm in radius, around a circle at constant speed 10 meters/second. What is the magnitude of the angular velocity?

Source Code:

```python
linear_velocity = 10  # in m/s
radius = 0.3  # in meters

angular_velocity = linear_velocity / radius
print(f"Angular velocity: {angular_velocity} rad/s")
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Angular velocity: 33.333333333333336 rad/s


Process finished with exit code 0
```

Question 4. A car with tires 50 cm in diameter travels 10 meters in 1 second. What is the angular speed?

Source Code:

```
linear_velocity = 10  # in m/s
radius = 0.25  # in meters

# Calculate angular speed
angular_speed = linear_velocity / radius
print(f"Angular speed: {angular_speed} rad/s")
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Angular speed: 40.0 rad/s

Process finished with exit code 0
```

Question 5. The angular speed of wheel 20 cm in radians is 120 rpm. What is the distance if the car travels in 10 seconds.

Source Code:

```
radius = 0.2  # in meters
angular_speed = 12.56  # in rad/s
time = 10  # in seconds
# Calculate linear speed
linear_speed = radius * angular_speed
# Calculate distance traveled
distance_traveled = linear_speed * time
print(f"Distance traveled in 10 seconds: {distance_traveled} meters")
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Distance traveled in 10 seconds: 25.120000000000005 meters

Process finished with exit code 0
```

Question 6: A car is running at a velocity of 50 miles per hour and the driver accelerates the car by 10 miles/hr^2. How far the car travels from this point in the next 2 hours, if the acceleration is constant. Formula: v = u + at

Source Code:

```python
initial_velocity = 50  # in miles/hour
acceleration = 10  # in miles/hour^2
time = 2  # in hours
# Step 1: Calculate final velocity
final_velocity = initial_velocity + (acceleration * time)
# Step 2: Calculate average velocity
average_velocity = (initial_velocity + final_velocity) / 2
# Step 3: Calculate distance traveled
distance_traveled = average_velocity * time

print(f"Distance traveled in 2 hours: {distance_traveled} miles")
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Distance traveled in 2 hours: 120.0 miles

Process finished with exit code 0
```

Question 7: A Stone is dropped freely from a height of 100 feet. With what velocity will it hit the ground? (Neglect the air resistance and assume the acceleration due to gravity is $32ft/s^2$). Formula: $v^2 - u^2 = 2as$

Source Code:

```python
import math
initial_velocity = 0  # in ft/s
acceleration = 32  # in ft/s^2
distance_fallen = 100  # in feet

# Using the formula v^2 - u^2 = 2as to calculate final velocity
final_velocity_squared = initial_velocity**2 + 2 * acceleration * distance_fallen
final_velocity = math.sqrt(final_velocity_squared)

print(f"Velocity at which the stone hits the ground: {final_velocity} ft/s")
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Velocity at which the stone hits the ground: 80.0 ft/s


Process finished with exit code 0
```